

Evidence: the screen test

The shadow-screen lab

Name _____ Date _____

Constructed observations. Same source, screen and sample position in A-C.

A • Opaque card

A clear, dark shape is seen on the screen.

C • Clear plastic

Most light passes through. A faint edge is visible.

B • Translucent paper

Some light passes through. The shape is lighter and less distinct.

D • Move the card away

The source stays on. The opaque card moves fully out of this beam path.

W1. Choose A, B or C for a clear dark silhouette. Give a reason.

W2. Predict what D changes on the screen. Explain.

Route 1 • Make a shadow plan

Word bank: source / card / screen / blocks.

S1. Which sample best makes a dark silhouette here?

A. A

B. B

C. C

S2. Complete: The opaque card ___ light.

S3. Put these in order: source / screen / card.

S4. Where do we see the shadow in this setup?

S5. Explain what happens when the card leaves the beam.

Route 2 • Write the design record

Use the observations, or add your own if a practical runs.

Task	Sample	What does the evidence show?
T1	A: opaque card	
T1	B: translucent paper	
T1	C: clear plastic	

T2. Draw source, card and screen. Mark where light is blocked and where the shadow is.

T3. Explain why “clear plastic can never cast a shadow” is too strong.

Route 3 • Repair the display explanation

A display says: “The black card paints a black copy of itself on the screen.”

X1. Replace this with a light-based explanation.

X2. Predict what a red opaque card of the same shape and size would do in the same position.

X3. Why might a shadow look grey rather than completely black in a classroom?

X4. Suggest a check that shows blocking, not coloured paint, causes the screen change.

Exit and reflection

Point, draw, write or tell an adult.

E1. What does an opaque card do to light?

E2. Where is the shadow in our setup?

E3. What changes when the card leaves the beam?

R1. Which idea did you improve after hearing evidence?

R2. What would you investigate or ask next?
